

# A Two-Component Solubility Model of the Onset of Asphaltene Flocculation in Crude Oils

J. X. Wang and J. S. Buckley\*

New Mexico Petroleum Recovery Research Center, New Mexico Institute of Mining and Technology, Socorro, New Mexico 87801

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A new, two-component asphaltene solubility model (ASM) has been developed to predict the phase behavior of asphaltenes in crude oils. Previous asphaltene solubility models have relied for solution upon simplifying assumptions (e.g., nucleation of a pure asphaltene or pure solvent phase) and approximation methods for determining crude oil solubility parameters. No such simplifying assumptions or approximations are needed if a simple correlation between the solubility parameter and the refractive index of nonpolar species is used to characterize the oil. Compositional ranges over which asphaltenes are stable, metastable, or unstable to phase separation can be defined from calculations of changes in Gibbs free energy using the Flory–Huggins theory for mixtures of crude oil and known amounts of hydrocarbon solvents and precipitants. Coupled values of the asphaltene solubility parameter and molar volume represent the only adjustable parameters in the model. Predictions have been tested against an extensive set of onset observations for crude oils and solutions of their asphaltenes. Experimental data consist of microscopic observations of the first appearance of asphaltene aggregates in response to addition of a minimum amount of a hydrocarbon precipitant (with carbon chain lengths varying from *n*-pentane to *n*-pentadecane). That first appearance of aggregate corresponds neither to the oil–precipitant mixture in which metastable conditions are first predicted nor to a mixture in which asphaltenes should be completely unstable. A consistent criterion between these two extremes has been defined as the “visible onset.” The proposed visible onset criterion has been successfully tested with a wide range of crude oils, precipitants, and solvents.

## Introduction

Problems associated with flocculation and deposition of asphaltenes can increase the costs of oil recovery and related processes. Preventing flocculation is, therefore, an important goal. Flocculation of asphaltene can be induced by changes in temperature, pressure, and composition that reduce the stability of asphaltenes in crude oils. Although experimental determination of the full set of thermodynamic conditions at which asphaltenes begin to flocculate might be desirable for each specific crude oil, the cost of such detailed characterization is prohibitive. An alternative is to obtain a minimum amount of laboratory data and incorporate that information into a thermodynamic model to predict the onset over a wide range of conditions. To mimic experimental observations, current thermodynamic models generally require the use of many adjustable parameters, some of which cannot be verified experimentally.<sup>1–3</sup>

While they can be tuned to match observed asphaltene behavior, they have limited predictive capability. Effective and practical methods to predict the onset of flocculation are needed.

## Asphaltene Solubility Models

Crude oil is such a complex mixture that describing the components of each oil in detail is impractical. With regard to asphaltene stability, the simplest approach is to treat crude oil as a mixture of two pseudo-components, with asphaltene as one component and the remainder of the oil as the other. If we further assume that asphaltene is solvated in its surrounding media, the well-known Flory–Huggins polymer theory,<sup>4,5</sup> which accounts for mixtures of very large molecules with much smaller solvent species, can be applied to calculate the solubility of asphaltene in the mixture. The first such solubility model for asphaltenes was proposed by Hirschberg et al.<sup>6</sup> Since then numerous applications and adaptations have been reported.<sup>7–13</sup>

\* Corresponding author. E-mail: jill@prrc.nmt.edu. Fax: 1–505-835-6031.

(1) Leontaritis, K. J.; Mansoori, G. A. Asphaltene Flocculation During Oil Production and Processing: A Thermodynamic Colloidal Model. In *Proceedings of the SPE International Symposium on Oilfield Chemistry*, San Antonio, Texas, 4–5 Feb 1987; Society of Petroleum Engineers: Richardson, Texas, 1987. SPE 16258.

(2) Nghiem, L. X.; Hassam, M. S.; Nutakki, R. Efficient Modeling of Asphaltene Precipitation. In *Proceedings of the 68th Annual Technical Conference and Exhibition of the Society of Petroleum Engineers*, Houston, Texas, 3–6 Oct 1993; Society of Petroleum Engineers: Richardson, Texas, 1993. SPE 26642.

(3) Pan, H.; Firoozabadi, A. A Thermodynamic Micellization Model for Asphaltene Precipitation: Part I: Micellar Size and Growth. In *Proceedings of the Annual Technical Conference and Exhibition of the Society of Petroleum Engineers*, Denver, 6–9 Oct 1996; Society of Petroleum Engineers: Richardson, Texas, 1996. SPE 36741.

(4) Flory, P. J. Thermodynamics of High Polymer Solutions. *J. Chem. Phys.* **1942**, *10*, 51.

(5) Huggins, M. L. Solutions of Long Chain Compounds. *J. Chem. Phys.* **1942**, *9*, 440.

According to Flory–Huggins theory, the Gibbs free energy change for asphaltene mixing with crude oil at constant pressure can be evaluated as

$$\Delta G_{\text{mix}} = RT \left[ n_m \ln \phi_m + n_a \ln \phi_a + n_m \phi_a \frac{v_m}{RT} (\delta_a - \delta_m)^2 \right] \quad (1)$$

where  $R$  is the universal gas constant,  $T$  is absolute temperature,  $n$  is number of moles,  $\phi$  is volume fraction,  $v$  is molar volume and  $\delta$  is the solubility parameter. Subscripts a and m represent asphaltene and the mixture of all nonasphaltene components (maltenes and any added hydrocarbon solvents and precipitants), respectively. The first two terms in eq 1 represent the Gibbs free energy change contributed by the combinatorial entropy change of mixing, while the third term arises from the enthalpy change upon mixing.

The chemical potentials for mixed solvent  $\Delta\mu_m$  and asphaltene  $\Delta\mu_a$  in the solution are the partial molar Gibbs free energies at constant temperature and pressure. Differentiating eq 1 results in eqs 2 and 3.<sup>6,11,14</sup>

$$\Delta\mu_m = RT \left[ \ln \phi_m + \phi_a \left( 1 - \frac{v_m}{v_a} \right) + \phi_a^2 \frac{v_m}{RT} (\delta_a - \delta_m)^2 \right] \quad (2)$$

$$\Delta\mu_a = RT \left[ \ln \phi_a + \phi_m \left( 1 - \frac{v_a}{v_m} \right) + \phi_m^2 \frac{v_a}{RT} (\delta_a - \delta_m)^2 \right] \quad (3)$$

As asphaltene solubility is reduced (by the addition of asphaltene precipitant in this work), phase separation can occur. Some asphaltenes remain in an asphaltene-poor lighter phase (phase L) while others concentrate in a heavier, asphaltene-rich phase (phase H). At phase equilibrium, the chemical potentials for each pseudo-component (i.e., asphaltene and mixed solvent) must be the same in both phases:

$$\Delta\mu_m^L = \Delta\mu_m^H \quad (4)$$

$$\Delta\mu_a^L = \Delta\mu_a^H \quad (5)$$

(6) Hirschberg, A.; deJong, L. N. J.; Schipper, B. A.; Meijer, J. G. Influence of Temperature and Pressure on Asphaltene Flocculation. *Soc. Pet. Eng. J.* **1984**, 24 (3), 283–293.

(7) Burke, N. E.; Hobbs, R. D.; Kashou, S. F. Measurement and Modeling of Asphaltene Precipitation. *J. Pet. Technol.* **1990**, 42 (11), 1440–1446.

(8) Novosad, Z.; Costain, T. G. Experimental and Modeling Studies of Asphaltene Equilibria for a Reservoir Under CO<sub>2</sub> Injection. In *Proceedings of the 65<sup>th</sup> Annual Technical Conference and Exhibition of the Society of Petroleum Engineers*, New Orleans, LA, 23–26 Sept 1990; Society of Petroleum Engineers: Richardson, Texas, 1990. SPE 20530.

(9) Nor-Azian, N.; Adewumi, M. A. Development of Asphaltene Phase Equilibria Predictive Model. In *Proceedings of the Eastern Regional Conference and Exhibition of the Society of Petroleum Engineers*, Pittsburgh, PA, 2–4 Nov 1993; Society of Petroleum Engineers: Richardson, Texas, 1993. SPE 26905.

(10) MacMillan, D. J.; Tackett, J. E.; Jesse, M. A.; Monger-McClure, T. G. A Unified Approach to Asphaltene Precipitation: Laboratory Measurement and Modeling. In *Proceedings of the Society of Petroleum Engineers International Symposium on Oilfield Chemistry*, San Antonio, Texas, 14–17 Feb 1995; Society of Petroleum Engineers: Richardson, Texas, 1995. SPE 28990.

(11) Cimino, R.; Corra, S.; Del Bianco, A.; Lockhart, T. P. Solubility and Phase Behavior of Asphaltenes in Hydrocarbon Media. In *Asphaltenes: Fundamentals and Applications*; Sheu, E. Y., Mullins, O. C., Eds.; Plenum Press: New York, 1995; pp 97–130.

(12) Rassamdana, H.; Dabir, B.; Nematy, M.; Farhani, F.; Sahimi, M. Asphalt Flocculation and Deposition: I. The Onset of Precipitation. *AIChE J.* **1996**, 42, pp 10–22.

Superscripts L and H represent phases L and H, respectively. Note that  $v_m$  and  $\delta_m$  in eqs 2 and 3 are functions of amount of precipitant added, i.e.,

$$v_m = \frac{\phi_p + \phi_s}{\phi_p/v_p + \phi_s/v_s} \quad (6)$$

$$\delta_m = \frac{\phi_p \delta_p + \phi_s \delta_s}{\phi_p + \phi_s} \quad (7)$$

with

$$\phi_m = \phi_p + \phi_s \quad \text{and} \quad \phi_m + \phi_a = 1 \quad (8)$$

where subscripts p and s represent precipitant and solvent (i.e., all nonasphaltene oil components as well as any added solvent species such as toluene). Combining eqs 2–5 produces nonlinear eqs 9 and 10, which can be solved numerically together with eqs 6–8 for given volumes of precipitant  $\phi_p$  to give asphaltene volume fractions in phase L ( $\phi_a^L$ ) and in phase H ( $\phi_a^H$ ) at phase equilibrium:<sup>14</sup>

$$\ln(1 - \phi_a^L) + \left( 1 - \frac{v_m}{v_a} \right) \phi_a^L + \chi (\phi_a^L)^2 = \ln(1 - \phi_a^H) + \left( 1 - \frac{v_m}{v_a} \right) \phi_a^H + \chi (\phi_a^H)^2 \quad (9)$$

$$\ln \phi_a^L + (1 - \phi_a^L) \left( 1 - \frac{v_a}{v_m} \right) + (1 - \phi_a^L)^2 \frac{v_a}{v_m} \chi = \ln \phi_a^H + (1 - \phi_a^H) \left( 1 - \frac{v_a}{v_m} \right) + (1 - \phi_a^H)^2 \frac{v_a}{v_m} \chi \quad (10)$$

where  $\chi$  is the Flory–Huggins interaction parameter

$$\chi = \frac{v_m}{RT} (\delta_a - \delta_m)^2 \quad (11)$$

The amount of asphaltene that would partition into each phase can be calculated based on conservation of mass.

**Pure Asphaltene Phase.** Hirschberg et al.<sup>6</sup> simplified the problem by assuming that the heavy phase H is pure asphaltene, i.e.,  $\Delta\mu_a^H = 0$ . This leads to

$$\ln \phi_a^L + (1 - \phi_a^L) \left( 1 - \frac{v_a}{v_m} \right) + (1 - \phi_a^L)^2 \frac{v_a}{RT} (\delta_a - \delta_m)^2 = 0 \quad (12)$$

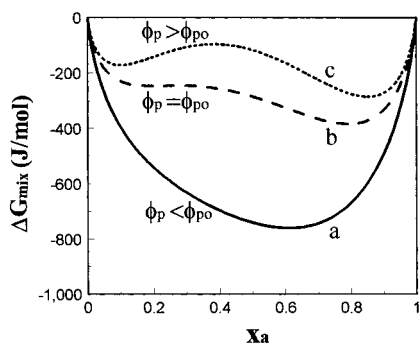
Equation 12 can be further simplified by assuming that  $\phi_a^L \ll 1$ , giving

$$\phi_a^L = \exp \left[ -1 + \frac{v_a}{v_m} - \frac{v_a}{RT} (\delta_a - \delta_m)^2 \right] \quad (13)$$

Equation 13 can be used to calculate the solubility of asphaltene in the solvent  $\phi_a^L$  for any composition. The onset of asphaltene precipitation is assumed to occur

(13) Yang, Z.; Ma, C.-F.; Lin, X.-S.; Yang, J.-T.; Guo, T.-M. Experimental and Modeling Studies on the Asphaltene Precipitation in Degassed and Gas-Injected Reservoir Oils. *Fluid Phase Equilib.* **1999**, 157, pp 143–158.

(14) Wang, J. X. Predicting Asphaltene Flocculation in Crude Oils. Ph.D. Thesis, New Mexico Institute of Mining and Technology, 2000.



**Figure 1.** Molar Gibbs free energy of mixing as a function of asphaltene mole fraction for three hypothetical mixtures of solvent, asphaltene, and precipitant (see Table 1). Phase separation can occur when the curve has two local minima. Curve a has less precipitant than the amount needed to initiate phase separation ( $\phi_p < \phi_{po}$ ). Curve c has more than the minimal amount of precipitant required for phase separation to occur ( $\phi_p > \phi_{po}$ ). Some mixture between these two corresponds to the onset of flocculation as defined by the first microscopic appearance of aggregated asphaltenes (e.g., curve b).

**Table 1. Parameters for Gibbs Free Energy Calculations**

|                                | solvent (s) | precipitant (p) | asphaltene (a) |
|--------------------------------|-------------|-----------------|----------------|
| $v$ (cm <sup>3</sup> /mol)     | 106.3       | 146.6           | 160.0          |
| $\delta$ (MPa <sup>1/2</sup> ) | 18.2        | 15.2            | 23.0           |

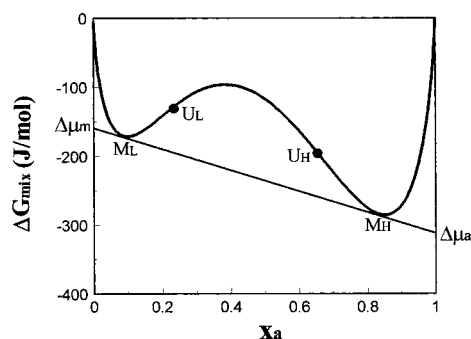
when the calculated solubility equals the asphaltene concentration in the mixture.

**Pure Solvent Phase.** Cimino et al.<sup>11</sup> assumed that, instead of a pure asphaltene phase, a pure solvent phase nucleates upon phase separation, i.e.,  $\Delta\mu_m^L = 0$ . This assumption leads to

$$\ln(1 - \phi_a^H) + \left(1 - \frac{v_m}{v_a}\right)\phi_a^H + (\phi_a^H)^2 \frac{v_m}{RT}(\delta_a - \delta_m)^2 = 0 \quad (14)$$

Equation 14 cannot be used to calculate the amount of asphaltene that separates. In fact, the pure solvent assumption implies that all asphaltenes are separated at the onset point. Instead, eq 14 can be used to calculate the amount of precipitant needed to trigger onset of precipitation. The volume fraction of asphaltene in the asphaltene-rich phase  $\phi_a^H$  is not a measurable quantity; it is treated as an adjustable parameter that is assumed to be about 0.8.

**Gibbs Free Energy of Mixing for Titration with Precipitant.** A typical titration test for asphaltenes begins with the asphaltenes dispersed in a good solvent (e.g., crude oil or toluene). To that dispersion, aliquots of precipitant (e.g., *n*-pentane or *n*-heptane) are added incrementally. The process is repeated until asphaltene aggregates are detected. To illustrate the relationships between  $\Delta G$  of mixing and the mole fraction of asphaltene ( $x_a$ ) for such a series of mixtures some arbitrary parameters have been selected (Table 1) and the resulting curves are plotted in Figure 1. In these calculations the amount of precipitant added increases from curve a to curve c. The case represented by curve a is a single phase. As more precipitant is added, the curves b and c develop two local minima. The emergence of a second local minimum is more subtle with asphaltenes, for which much higher values of  $v_a$  are expected, but the principle is the same as in this illustrative example.



**Figure 2.** Mixtures of the components defined in Table 1 with the volume fraction of precipitant equal to 1.5 times that of solvent so that  $\phi_p > \phi_{po}$ . The line with two points of tangency defines the chemical potentials of components a and m ( $m = s + p$ ), and the compositions and relative amounts of coexisting phases L and H. Points  $M_L$  and  $M_H$  are the phase compositions of L and H, respectively, and the lower and upper limits of metastability. Points  $U_L$  and  $U_H$  define the unstable region.

In Figure 2, the case with an excess of precipitant is examined. Constructing a line that has two points of tangency ( $M_L$  and  $M_H$ ) to the Gibbs free energy curve defines relative amounts and compositions of two coexisting phases. If the mole fraction of asphaltene  $x_a$  falls between two inflection points  $U_L$  and  $U_H$ , the mixture is unstable and an immediate phase split occurs into a lighter phase with composition  $M_L$  and a heavier phase with the composition  $M_H$ . Between  $M_L$  and  $U_L$  or  $M_H$  and  $U_H$ , mixtures are metastable and may separate slowly. Below  $M_L$  and above  $M_H$ , no phase split occurs.

**Onset Criteria.** As increasing amounts of precipitant are added, a family of  $\Delta G_{mix}$  vs  $x_a$  curves is generated. Which of these curves corresponds to the onset of asphaltene precipitation? The answer to that question depends to some extent on the experimental technique by which the onset is identified. Visibility of aggregates is limited by the wavelength of visible light; other techniques may identify smaller aggregates. If all the input parameters could be measured, calculations of Gibbs free energy of mixing at the onset of precipitation would produce the correct curve. Lacking the asphaltene parameters, some criterion for distinguishing the onset condition must be deduced.

Hirschberg et al.<sup>6</sup> used the amount of asphaltene in the onset mixture as the upper limit of solubility. The combination of asphaltene, solvents and precipitants that gives an asphaltene concentration that matches  $M_L$  is judged to be the onset mixture. Note, however, that  $M_L$  is shifted in this case by their assumption that  $x_a = 1$  in phase H, which gives a different composition for the asphaltene-poor phase at  $M_L$  than does the mutual tangent construction. Cimino et al.<sup>11</sup> decided which curve corresponds to the onset by setting an arbitrary value for the fraction of asphaltene in the asphaltene-rich phase,  $\phi_a^H \approx 0.8$ . Similarly, their calculation of the  $M_H$  point is biased to some extent by assuming  $x_a = 0$  in phase L. In this work we show that the visible onset corresponds well to curve b in Figure 1. The characteristic feature of curve b is that the tangent line through the inflection point at  $U_L$  is horizontal. Logical arguments could be made for any of these criteria. Differentiating between them requires evaluation of the predictions they make with respect to experimental

observations. Those observations and comparisons are treated in the following sections.

**Input Parameters.** There are four input parameters in eqs 9, 10, 13, and 14, i.e., the solubility parameters and molar volumes of each of the two pseudo-components ( $\delta_a$ ,  $\delta_m$ ,  $v_a$ , and  $v_m$ ). An average molar volume of the nonasphaltic mixture can be estimated from measurements of density and average molecular weight. There is no direct method for measurement for solubility parameters of complex mixtures, however, so both  $\delta_m$  and  $\delta_a$  are unknown. Estimation techniques for  $\delta_a$  involve observations of solubility of asphaltenes in compounds with known solubility parameters.<sup>7</sup> Solubility of onset mixtures has been estimated from onset data in very dilute solutions of asphaltenes or oil in solvent (usually toluene) and precipitant,<sup>11</sup> but the implicit assumption that onset at infinite dilution occurs at the same solubility conditions as the onset in oil is unlikely to be correct.<sup>15</sup> Since molecular weight is not well-defined for asphaltenes and density is estimated from dilute solution,  $v_a$  is always unknown.

A simple method for estimating the solubility parameter of mixtures of hydrocarbons makes use of the linear relationship between solubility parameter and a function of the readily measured quantity, refractive index (RI).<sup>16</sup> Figure 3 illustrates the relationship between  $\delta$  and  $F_{RI}$  (where  $F_{RI} = [(RI)^2 - 1]/[(RI)^2 + 2]$ ) for selected  $n$ -paraffins and aromatics.

Thus, the nonasphaltic hydrocarbon mixture properties  $v_m$  and  $\delta_m$  are determined by measurements. The remaining properties  $v_a$  and  $\delta_a$  are the only adjustable parameters in the model. Reducing the uncertainty in these input parameters eliminates the need for simplifying assumptions.

**Algorithm for Solution of the Full Asphaltene Solubility Model (ASM).** The solution algorithm is outlined in Figure 4. Since mixture properties vary as precipitant is added, properties of both the original oil and the precipitants are needed. Standard mixing rules, summarized in eqs 15–17, are used to calculate mixture properties:

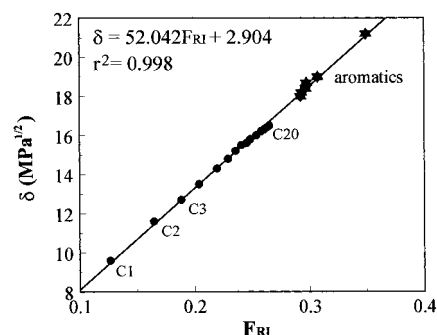
$$\frac{1}{v} = \sum_i \frac{\phi_i}{v_i} \quad (15)$$

$$\delta = \sum_i \phi_i \delta_i \quad (16)$$

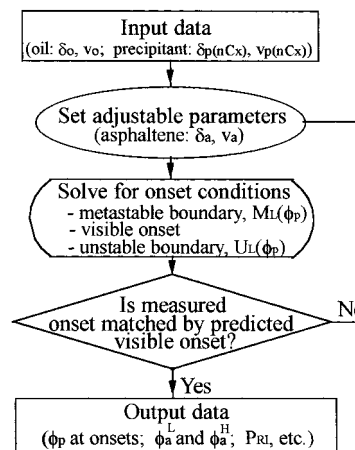
$$F_{RI} = \sum_i \phi_i F_{RI,i} \quad (17)$$

where the subscript  $i$  represents the  $i$ th component in the mixture.

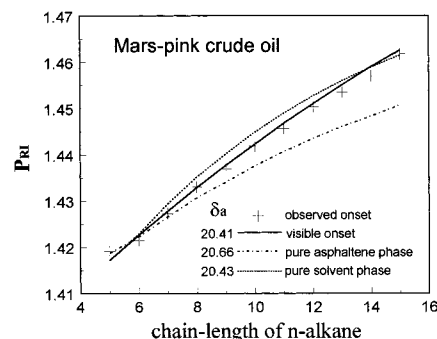
Numerical solutions of the ASM result in three key points with respect to asphaltene stability: the metastable boundary at which the potential for asphaltene flocculation exists, the visible onset point where as-



**Figure 3.** Linear correlation between solubility parameter and  $F_{RI}$  (data adapted from refs 17 and 18).



**Figure 4.** Solution algorithm flowchart.



**Figure 5.** Experimental observations of the onset conditions for asphaltene precipitation from Mars-Pink crude oil. Precipitation was induced by mixing oil with  $n$ -alkanes of varying carbon chain length from pentane to pentadecane. Fits to the experimental data were optimized for each of the three solubility models discussed in the text. Input parameters:  $v_o = 319 \text{ cm}^3/\text{mol}$ ,  $\delta_o = 19.25 \text{ MPa}^{1/2}$ ,  $v_a = 2500 \text{ cm}^3/\text{mol}$ .

phaltenes can first be observed microscopically, and the absolute instability point at which flocculation must occur. Asphaltene will not flocculate if solution conditions are in the region beyond the metastable boundary.

**Example: Onset of Asphaltene Precipitation from Mars-Pink Crude Oil.** Application of ASM can best be illustrated by example. Samples of Mars-Pink crude oil were mixed with varying amounts of  $n$ -alkane precipitants from  $n$ -pentane to  $n$ -pentadecane to determine the onset conditions for each. Refractive index of the mixture at the onset ( $P_{RI}$ ) was recorded for each precipitant, as shown in Figure 5. The advantages of using  $P_{RI}$  to characterize the onset conditions are discussed by Buckley et al.<sup>16</sup> Also shown in Figure 5

(15) Wang, J. X.; Buckley, J. S. An Experimental Approach to Prediction of Asphaltene Flocculation. In *Proceedings of the Society of Petroleum Engineers International Symposium on Oilfield Chemistry*, Houston, Texas, 13–16 Feb 2001; Society of Petroleum Engineers: Richardson, Texas, 2001. SPE 64994.

(16) Buckley, J. S.; Hirasaki, G. J.; Liu, Y.; Von Drasek, S.; Wang, J. X.; Gill, B. S. Asphaltene Precipitation and Solvent Properties of Crude Oils. *Pet. Sci. Technol.* **1998**, 16 (3/4), pp 251–285.

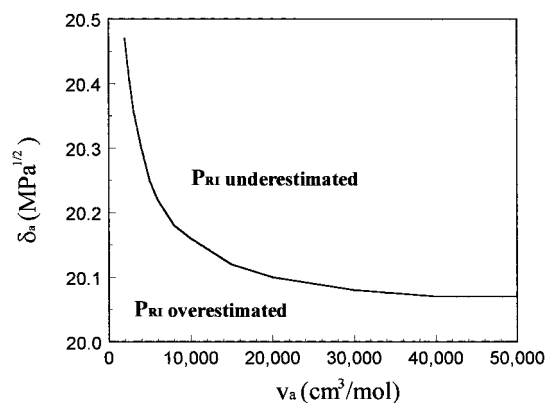


**Table 2. Comparisons for Three Models Based on Flory–Huggins Polymer Theory**

| characteristics                                          | pure asphaltene phase<br>(Hirschberg et al. <sup>6</sup> )                                      | pure solvent phase<br>(Cimino et al. <sup>11</sup> )                                                                   | ASM                                                                                                |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| significant assumptions                                  | pure solid asphaltene phase nucleating at phase separation                                      | pure solvent phase nucleating at phase separation, while asphaltene forms an asphaltene-rich phase                     | phase separation creates asphaltene-rich and asphaltene-poor phases                                |
| input parameters                                         | $v_m, v_a$<br>$\delta_m, \delta_a$<br>wt % <i>n</i> -C <sub>7</sub> asphaltene<br>T             | $v_m, v_a$<br>$\delta_m, \delta_a$<br>$\phi_a^H$<br>T                                                                  | $v_m, v_a$<br>$\delta_m, \delta_a$<br>wt % <i>n</i> -C <sub>7</sub> asphaltene<br>T                |
| onset criteria                                           | $\phi_a^L$ equals $\phi_a$ in onset mixture                                                     | onset is determined by setting a constant asphaltene concentration in asphaltene-rich phase ( $\phi_a^H \approx 0.8$ ) | onset is determined by horizontal tangent to inflection point ( $U_L$ ) on Gibbs free energy curve |
| sensitivity of calculated onset on asphaltene properties | very sensitive to wt % <i>n</i> -C <sub>7</sub> asphaltene and $\delta_a$<br>sensitive to $v_a$ | very sensitive to $\delta_a$<br>not sensitive to $v_a$ or wt % <i>n</i> -C <sub>7</sub> asphaltene                     | very sensitive to $\delta_a$<br>not sensitive to $v_a$ or wt % <i>n</i> -C <sub>7</sub> asphaltene |

are three fits to the data. In all cases,  $v_a$  was assumed to be 2500 cm<sup>3</sup>/mol. Values of the asphaltene solubility parameter are shown in the legend. Solubility parameters of the oil and oil plus *n*-alkane at the onset conditions are based on measured refractive indices. The pure asphaltene assumption of Hirschberg et al.<sup>6</sup> can be made to reproduce any one of the data points by changing the input value of  $\delta_a$ . In Figure 5, only the lowest molecular weight precipitants fit the predicted curve for the parameters chosen. While the prediction can be shifted vertically by changing the input parameters, no combination of  $v_a$  and  $\delta_a$  gives the observed slope of  $P_{RI}$  vs precipitant carbon chain length. The assumption of a pure solvent phase comes closer to describing the experimental observations, but ASM and visible onset criterion described above, without further simplifying assumptions, correlates the full range of data at least as well. Although the arguments for various onset criteria may have been made in other terms, the underlying justification for both the value of  $\phi_a^H \approx 0.8$  assumed in the pure solvent model and the visible onset criterion in ASM is that they correctly predict the slope in Figure 5. A summary of the comparisons between the three models is presented in Table 2.

Although both  $\delta_a$  and  $v_a$  are adjustable parameters in ASM, there are some logical constraints on their values. As shown in Figure 6,  $\delta_a$  and  $v_a$  are coupled; together they produce a family of solutions, all of which match the experimental data for Mars-Pink crude oil. It is logical, therefore, to fix one of the values and use the other as the sole fitting parameter. If  $v_a$  is selected as the fixed parameter, Figure 6 shows that values above about 10 000 cm<sup>3</sup>/mol would reduce the sensitivity to  $\delta_a$ . While values well in excess of 10 000 cm<sup>3</sup>/mol might be deduced from the literature,<sup>19</sup> more recent estimates tend to be much lower.<sup>20,21</sup> In selecting a value at which to fix  $v_a$ , it is also necessary to ensure that

**Figure 6.** Pairs of adjustable parameters ( $\delta_a$  and  $v_a$ ) that match experimental data for Mars-Pink precipitation (cf. Figure 5).

predictions are physically reasonable. If the value of  $v_a$  is fixed at too low a value (1500 cm<sup>3</sup>/mol in this example), predictions of the amount of asphaltene in the asphaltene-poor phase at onset are clearly too high, as they can exceed the total amount of *n*-heptane asphaltenes in Mars-Pink crude oil for precipitants larger than *n*-nonane, as shown in Figure 7. The value of 2500 cm<sup>3</sup>/mol used for the comparisons reported here is a reasonable compromise between these two constraints.

There is, of course, some degree of uncertainty in all the input properties, those that are measured as well as those that are treated as adjustable parameters. It is important, therefore, to evaluate the effect of uncertainty on model predictions. Using the data for Mars-Pink, the effects of varying  $v_a$ ,  $\delta_a$ ,  $v_o$ , and wt % of *n*-heptane asphaltenes in the original oil were all examined (Figure 8). Of these parameters, the greatest sensitivity was to the value of asphaltene solubility parameter (Figure 8b); the onset calculation was least sensitive to the amount of *n*-heptane asphaltene in the

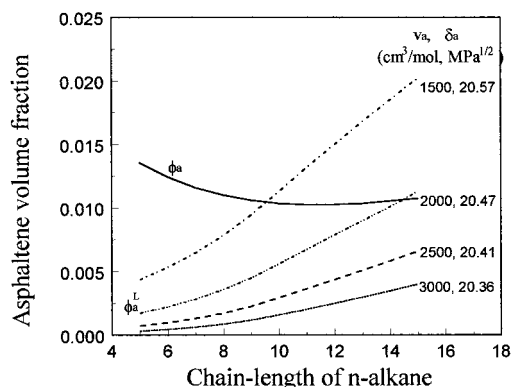
(17) Barton, A. F. M. *CRC Handbook of Solubility Parameters and Other Cohesion Parameters*, 2nd ed.; CRC Press: Boca Raton, 1991.

(18) Weast, R. C. *CRC Handbook of Chemistry and Physics*, 68th ed.; CRC Press: Boca Raton, 1987.

(19) Speight, J. G.; Wernick, D. L.; Gould, K. A.; Overfield, R. E.; Rao, B. M. L.; Savage, D. W. Molecular Weight and Association of Asphaltenes: A Critical Review. *Revue de l'Institut Français du Pétrole* **1985**, 40 (1), 51–61.

(20) Speight, J. G.; Plancher, H. Molecular Models for Petroleum Asphaltenes and Implications for Asphalt Science and Technology. In *Proceedings of the International Symposium on the Chemistry of Bitumens*; Rome, 1991; p 154.

(21) Groenzin, H.; Mullins, O. C. Molecular Size and Structure of Asphaltenes from Various Sources. *Energy & Fuels* **2000**, 14, 677–684.



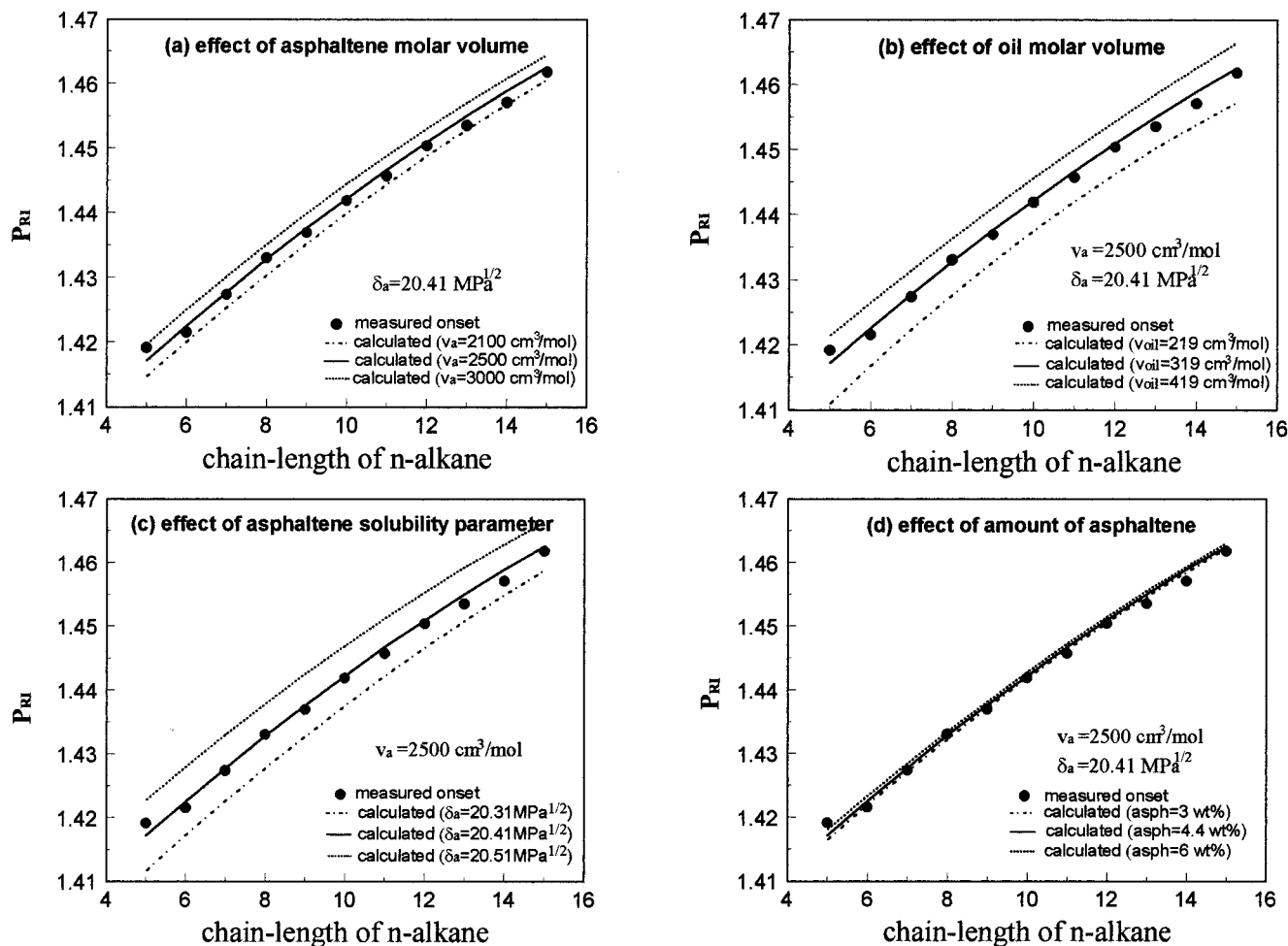
**Figure 7.** Calculated values of the volume fraction of asphaltene in the lighter (asphaltene-poor) phase at the onset ( $\phi_a^L$ , dashed lines) can exceed the amount of asphaltene in the onset mixture ( $\phi_a$ , solid line) for low values of  $v_a$ , illustrated here for Mars-Pink crude oil. Thus, there is a lower limit on values of  $v_a$  that give physically meaningful results. (Similarly, there is an upper limit on  $\delta_a$ .)

oil sample (Figure 8d). Asphaltene density is not well established, but essentially no differences in predictions were found for assumed values from 1 to 1.2 g/cm<sup>3</sup>, the range in which asphaltene densities are usually estimated. Perhaps the most important point that should be emphasized is that none of the input parameters tested had any appreciable effect on the predicted slope of  $P_{RI}$  vs  $n$ -alkane chain length. Only changes in the onset criterion alter this slope.

## Comparisons with Experimental Data

**Precipitation from Crude Oils.** ASM produces excellent matches to experimental data for the onset of asphaltene precipitation from many crude oils, not just Mars-Pink. Table 3 summarizes physical properties of seven crude oils. Five of the oils, A-93, A-95, Lagrave, Moutray, and Tensleep, had some asphaltene aggregates visible in the samples as received. A good asphaltene solvent (1-methylnaphthalene, abbreviated as 1-MN) was added to those oils to disperse existing aggregates. The amounts added are shown in Table 3. Details of the experimental measurements have been reported elsewhere.<sup>14,15</sup> The experimental onset data and the ASM fits are shown in Figure 9; the values of  $\delta_a$  for these fits are included in Table 3 in the column labeled  $\delta_a$  (crude oil). The value of  $v_a$  was fixed at 2500 cm<sup>3</sup>/mol for all oils. Asphaltene solubility parameters range from a low of 20.32 for Lost Hills (most stable) to a high of 21.28 (least stable) for Lagrave, an oil known for asphaltene problems.<sup>22</sup> ASM gives acceptable fits to all of the experimental data.

**Addition of Solvents.** Solvents have been added to some crude oil samples to simplify the task of identifying the asphaltene aggregates that form in response to changes in composition. In many dead oil samples, a variety of particles including precipitated asphaltenes can be observed microscopically. Addition of 1-MN, a good asphaltene solvent, can reduce interference from preexisting aggregates. This approach raises questions

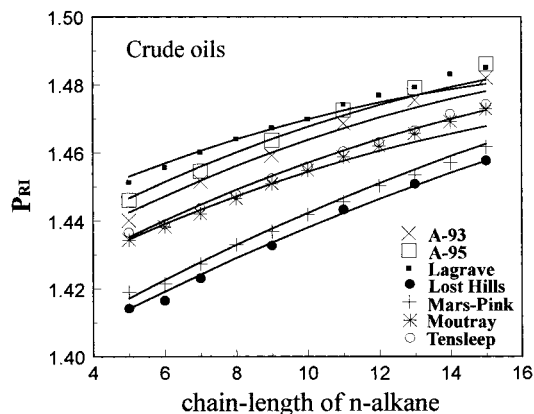


**Figure 8.** Sensitivity of ASM model predictions to uncertainty in input variables illustrated for Mars-Pink crude oil.

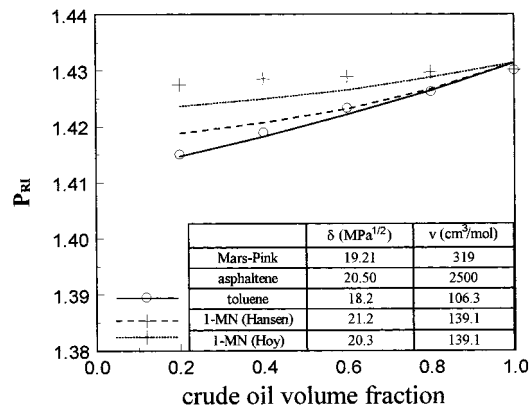
Table 3. Properties of Crude Oils and Asphaltenes at 20 °C and 1 Atm Pressure

| crude oil              | <i>n</i> -C <sub>7</sub> asphalt<br>(wt %) | RI     | $\rho$ (g/cm <sup>3</sup> ) | $v_o$ (cm <sup>3</sup> /mol) | 1-MN added (%) | $v_a$ (cm <sup>3</sup> /mol) | $\delta_a$ (crude oil)<br>(MPa <sup>1/2</sup> ) | $\delta_a$ ( <i>n</i> -C <sub>7</sub> asphalt)<br>(MPa <sup>1/2</sup> ) |
|------------------------|--------------------------------------------|--------|-----------------------------|------------------------------|----------------|------------------------------|-------------------------------------------------|-------------------------------------------------------------------------|
| A-93                   | 6.87                                       | 1.5220 | 0.8911                      | 287                          | 30             | 2500                         | 20.60                                           | 21.15                                                                   |
| A-95                   | 8.67                                       | 1.5159 | 0.8879                      | 305                          | 30             | 2500                         | 20.80                                           | 21.40                                                                   |
| Lagrove                | 6.69                                       | 1.4832 | 0.8136                      | 231                          | 30             | 2500                         | 21.28                                           | 21.64                                                                   |
| Lost Hills             | 2.78                                       | 1.5137 | 0.9173                      | 289                          | 0              | 2500                         | 20.32                                           |                                                                         |
| Mars-Pink <sup>a</sup> | 4.40                                       | 1.5380 | 0.9514                      | 319                          | 0              | 2500                         | 20.41 <sup>a</sup>                              | 21.25                                                                   |
| Moutray                | 1.29                                       | 1.4827 | 0.8488                      | 285                          | 20             | 2500                         | 20.78                                           | 21.15                                                                   |
| Tensleep               | 3.20                                       | 1.4880 | 0.8685                      | 317                          | 35             | 2500                         | 20.92                                           | 21.90                                                                   |

<sup>a</sup> In later tests (shown in Figures 10 and 11) the value of  $P_{RI}$  was 1.4301, giving  $\delta_a = 20.5 \text{ MPa}^{1/2}$



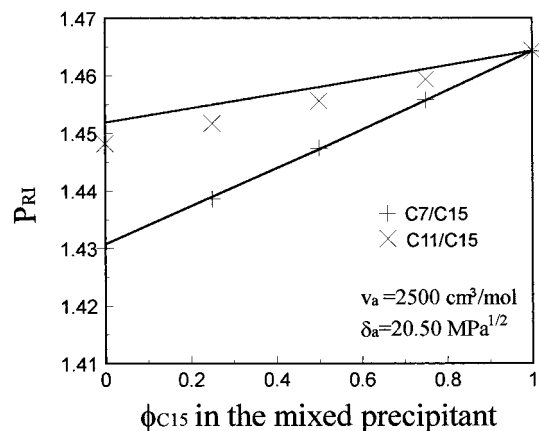
**Figure 9.** Comparison of experimental and calculated onset conditions for six crude oils. Precipitants were *n*-alkanes from pentane to pentadecane. In each case,  $v_a = 2500 \text{ cm}^3/\text{mol}$ . Other input parameters are included in Table 3.



**Figure 10.** Addition of toluene changes the onset of precipitation from Mars-Pink crude oil induced by *n*-heptane as predicted by ASM. Addition of 1-MN has little effect on the onset conditions, less expected from the model predictions. Two values of solubility parameter for 1-MN have been found in the literature (Barton, 1991). Of these, the Hoy parameter comes closer to predicting the onset experiments. Symbols represent measured  $P_{RI}$ , while lines represent each corresponding model prediction.

about the effect of solvent addition on the observed and predicted onset values. To investigate these questions, a crude oil that does not have preexisting asphaltene aggregate particles, such as Mars-Pink, is required.

Figure 10 shows experimental results of addition of varying amounts of toluene and 1-MN to the Mars-Pink oil. Also shown are the ASM predictions. In this series



**Figure 11.** Asphaltene precipitation onset conditions induced by mixtures of *n*-alkane precipitants from Mars-Pink crude oil varies almost linearly with the component volume fractions and is well predicted by ASM.

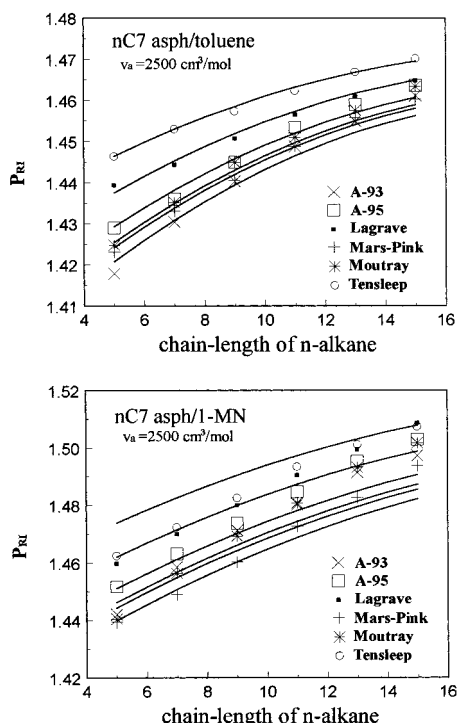
of experiments, the neat crude oil had a slightly different onset point with *n*-heptane than that shown in previous figures, presumably due to changes in the oil sample with time. Thus, the solubility parameter quoted for Mars-Pink in Figure 10 is slightly higher than that in Table 3. The ASM model takes into account changes in oil solubility parameter and molar volume as solvent is added and the predictions are revised accordingly.

Mixtures with toluene show a steadily decreasing value of  $P_{RI}$  as the amount of toluene in the mixture increases, a change that is also predicted by the ASM model. Little change in  $P_{RI}$  is observed upon addition of 1-MN over a wide range of volume fractions, an observation that is encouraging since addition of moderate amounts of 1-MN is helpful experimentally. The model, however, predicts that the onset values should decrease, although not as much as the change brought about by the addition of toluene. Poor correspondence between experimental results and the ASM model may be partly accounted for by uncertainty in the 1-MN solubility parameter. Two values are reported by Barton,<sup>17</sup> as shown in Figure 10. The lower value comes much closer to matching the experimental data than does the higher value.

**Mixtures of Precipitants.** In situations of practical interest, it is the mixture of paraffins in a crude oil that can, with changes in pressure and temperature, cause separation of an asphaltene phase. ASM has been shown to predict accurately the onset of asphaltene precipitation induced by mixtures of *n*-paraffins as shown in Figure 11 for Mars-Pink crude oil.

**Heptane Asphaltenes Reprecipitated from Solutions.** Often asphaltene studies are made using solutions of precipitated solid asphaltenes dissolved in

(22) Garland, E. The Asphaltic Properties of an Apparently Ordinary Crude Oil May Lead to Re-Thinking of Field Exploitation. In *Proceedings of the Annual Technical Conference and Exhibition of the Society of Petroleum Engineers*, San Antonio, Texas, 8–11 Oct 1989; Society of Petroleum Engineers: Richardson, Texas, 1989. SPE 19731.



**Figure 12.** ASM gives excellent fits to observations of reprecipitation of *n*-heptane asphaltenes from toluene solution. The solubility parameters are summarized in Table 3. Using those values of  $\delta_a$  to predict the onset from solutions in 1-MN produces matches to some of the data, but the slopes of the predicted curves are different than experimental observations: (a) 1% *n*-heptane asphaltenes in toluene, (b) 1% *n*-heptane asphaltenes in 1-MN.

toluene or some other asphaltene solvent. Asphaltenes, precipitated by addition of *n*-heptane, were recovered from six of the crude oils listed in Table 3 and 1% solutions were prepared in either toluene or 1-MN. The onset observations and best fits produced by ASM are in very good agreement for the solutions in toluene (Figure 12a). The values of the asphaltene solubility parameter for these fits are included in Table 3. Using these solubility parameters, predictions can be made for the same asphaltenes dissolved in 1-MN, as shown in Figure 12b. The predictions fit some of the observations, either for the lower or higher molecular weight precipitants. However, the predicted slopes are consistently lower than those observed, suggesting a failure of the onset criterion in the 1-MN solutions. Whether this reflects a limitation of the model or experimental problems remains to be determined.

Asphaltenes dissolved in toluene reproduce neither the solubility conditions at the onset (compare Figure 12a with Figure 9) nor the asphaltene solubility parameter (Table 3) of the crude oil from which they were obtained. Studies that focus on the behavior of separated asphaltenes or that require high dilutions of oil may well be misleading about the stability of asphaltenes in the original oil.

### Summary

Evaluation of the potential for separation of an asphaltene phase from crude oil would be greatly simplified if the full range of unstable conditions could be extrapolated from a few simple measurements. For

most of the oils tested and for their asphaltenes in toluene solution, ASM provides a reliable means to predict onset conditions, as defined by  $P_{RI}$ , for a wide range of precipitants. The improved predictive capability of the model over earlier solubility models comes from three key innovations:

- estimation of solubility parameters based on RI measurements;
- full solution of the thermodynamic equations to give compositions of both asphaltene-poor and asphaltene-rich phases; and
- use of the Gibbs free energy curve to define onset conditions that do not depend on the amount of asphaltene in the original oil sample.

**Limitations of the Two-Component Model.** The correspondence between theoretical prediction provided by ASM and experimental data is surprisingly good, considering the complexity of the crude oils that are being defined by only two components. In most cases, the same solubility parameter can describe the conditions under which asphaltenes are precipitated by pentane and pentadecane, despite the fact that the amount and composition of material precipitated by excess amounts of these two alkanes would be considerably different. Previous studies have found that close to the onset, the higher molecular weight species are those that preferentially separate into the asphaltene-rich phase.<sup>23,24</sup> Thus, the initially separating species may be more similar than the average of all species that are insoluble in an excess of different precipitants would be. It should be expected, therefore, that the two-component model would be less able to predict the amount of asphaltene produced by different precipitants. For this, more complex approaches will be required, but for prediction of the very first appearance of asphaltenes, the two-component ASM approach appears likely to be adequate. To extend onset predictions to conditions of practical interest, measurements of RI as a function of temperature, pressure, and composition are needed.

### Conclusions

- A thermodynamic model has been rigorously derived from Flory–Huggins polymer theory, without any arbitrary assumptions about the compositions of the two phases formed when the asphaltenes separate from the oil. In principle, this asphaltene solubility model (ASM) is fully predictive once parameters are established for asphaltene properties.

- Numerical solutions of the ASM result in three points with respect to asphaltene stability: the metastable boundary at which the potential for asphaltene flocculation exists, the visible onset point where asphaltenes can first be observed microscopically, and the absolute instability point at which flocculation must occur. Asphaltene will not flocculate if solution conditions are in the region beyond the metastable boundary.

(23) Fuhr, B. J.; Cathrea, C.; Coates, L.; Kalra, H.; Majeed, A. I. Properties of Asphaltenes from a Waxy Crude. *Fuel* **1991**, 70, pp 1293–1297.

(24) Andersen, S. I.; Stenby, E. H. Precipitation of Asphaltenes in Mixed Solvents. In *Proceedings of the Third International Symposium on Reservoir Wettability and Its Effect on Oil Recovery*; University of Wyoming: Laramie, WY, 1996; pp 59–62.



- A lower limit exists for asphaltene molar volume below which no realistic prediction for onset of asphaltene flocculation could be obtained. Too high a value of asphaltene molar volume, on the other hand, will result in a wider transition zone between metastable and visible onset conditions.

- The ASM is able to reproduce a wide range of experimental observation of the onset of asphaltene flocculation.

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### Nomenclature

$F_{RI}$ : a function of RI,  $F_{RI} = [(RI)^2 - 1]/[(RI)^2 + 2]$  (dimensionless)

$n$ : number of mole (mol)

$P_{RI}$ : refractive index of mixture at onset of asphaltene flocculation (dimensionless)

$r$ : correlation coefficient (dimensionless)

$R$ : universal gas constant (8.31441 J/K/mol)

RI: refractive index (dimensionless)

$T$ : absolute temperature (K)

$v$ : molar volume ( $\text{cm}^3/\text{mol}$ )

wt %: weight percentage (dimensionless)

$x$ : molar fraction (dimensionless)

$\chi$ : Flory–Huggins interaction parameter (dimensionless)

$\delta$ : Hildebrand solubility parameter ( $\text{MPa}^{1/2}$ )

$\phi$ : volume fraction (dimensionless)

$\Delta G_{\text{mix}}$ : change of Gibbs free energy upon mixing (J)

$\Delta\mu$ : chemical potential (J/mol)

$\rho$ : density ( $\text{g}/\text{cm}^3$ )

### Subscripts

m: remainder of the mixture except asphaltene

a: asphaltene

o: crude oil

p: asphaltene precipitant

s: asphaltene solvent

### Superscripts

L: asphaltene-poor lighter phase formed when phase separation occurs

H: asphaltene-rich heavier phase formed when phase separation occurs

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